# Computational Replication Report: Chatterjee, Ghosh, Nandy & Saha (2023/24), *Second-order topological superconductor via noncollinear magnetic texture* (arXiv:2308.12703)

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## 1 Paper summary

The paper proposes a route to a two-dimensional *second-order topological superconductor* (SOTSC) hosting Majorana corner modes (MCMs) by sandwiching a 2D quantum spin Hall insulator (QSHI, BHZ model) between a noncollinear spin-spiral magnetic texture and an $s$-wave superconductor — crucially *without* Rashba SOC or an external field. The noncollinear texture supplies, via a unitary transformation, both an effective in-plane Zeeman field and an effective spin-orbit coupling that together stabilize the higher-order phase. The bulk topology is characterized by a quantized quadrupole moment $Q_{xy} = 1/2$, and the phase is diagnosed numerically by four zero-energy corner-localized states.

## 2 Scope statement

We implement the **exact real-space BdG lattice model** (Eq. 1) and test its three headline claims directly. The effective low-energy continuum model (Eq. 4), its analytic derivation, the edge theory, and the analytic phase boundary (white lines in Fig. 3) are *not* reproduced; they are analytical supplements, out of scope for a numerical replication. Lattice sizes are reduced for CPU tractability ($L = 24$ for MCM counting via sparse shift-invert; $L = 14$ for the dense many-body quadrupole) versus the paper's $30 \times 30$.

## 3 Model (Eq. 1)

The $8 \times 8$ BdG Hamiltonian on an $L_x \times L_y$ square lattice:

$$H = \sum_{i,j} c^\dagger_{i,j} \big[ \{ \epsilon_0 \Gamma_1 + \Delta_0 \Gamma_2 + J_{\rm ex}(\Gamma_3 \cos\phi_{ij} + \Gamma_4 \sin\phi_{ij}) \} c_{i,j} - \{ t\Gamma_1 + i\lambda_x \Gamma_5 \} c_{i+1,j} - \{ t\Gamma_1 + i\lambda_y \Gamma_6 \} c_{i,j+1} \big] + \text{h.c.},$$

with $\Gamma_1 = \tau_z\sigma_z s_0$, $\Gamma_2 = \tau_x\sigma_0 s_0$, $\Gamma_3 = \tau_0\sigma_0 s_x$, $\Gamma_4 = \tau_0\sigma_0 s_y$, $\Gamma_5 = \tau_z\sigma_x s_z$, $\Gamma_6 = \tau_z\sigma_y s_0$ ($\tau$=particle-hole, $\sigma$=orbital, $s$=spin). Spiral: $\phi_{ij} = g_x x + g_y y$, with $g_x = g_y = g$, $\lambda_x = \lambda_y = \lambda$. Fig.-2 parameters: $t = 1$, $J_{\rm ex} = 0.8$, $g = 0.2$, $\Delta_0 = 0.4$, $\lambda = 0.5$, $\epsilon_0 = 1$.

## 4 Claims table

| ID | Claim | Type | Testable? | Tested? |
|---|---|---|---|---|
| C1 | 4 zero-energy corner modes (MCMs) in topological phase | numerical | yes | **yes** |
| C2 | Bulk quadrupole $Q_{xy} = 1/2$ (topo), 0 (trivial) | numerical | yes | **yes** |
| C3 | Topological→trivial transition with increasing $g$ | numerical | yes | **yes** |
| C4 | Effective model Eq. 4 (Zeeman + SOC from texture) | analytic | partial | no |
| C5 | Low-energy edge theory / analytic phase boundary | analytic | partial | no |

## 5 Method

Python 3.14, NumPy, SciPy (sparse `eigsh` shift-invert around $E = 0$ for MCM counts; dense `eigh` for the occupied-manifold quadrupole). The many-body quadrupole (Eq. 2) uses the standard Kang–Fang–Fu / Wheeler–Wang–Uchida formula $Q_{xy} = \frac{1}{2\pi}\mathrm{Im}\ln\det(U^\dagger W U) - \langle q_{xy}\rangle_{\mathrm{occ}} \pmod 1$, with $W = \exp(i2\pi\,\hat{x}\hat{y}/L_x L_y)$ and $U$ the occupied (negative-energy) BdG eigenvectors. Run: `python3 code/chatterjee2023_replication.py`.

## 6 Results vs paper

| Quantity | Paper | This work | Match |
|---|---|---|---|
| # zero modes (topo, OBC) | 4 | **4** | ✓ |
| zero-mode energy | $\sim 10^{-7}$ | $\sim 1.3 \times 10^{-4}$ (finite size) | ✓(qual.) |
| corner localization of 4 modes | 4 corners | 92.5% in corner boxes | ✓ |
| $Q_{xy}$ (topological, $g = 0.2$) | 0.5 | **0.5000** | ✓ |
| $Q_{xy}$ (trivial, $g = 1.4$) | 0 | **0.0000** | ✓ |
| transition in $g$ | yes | 4 modes for $g \leq 0.7$, 0 by $g = 1.0$ | ✓ |

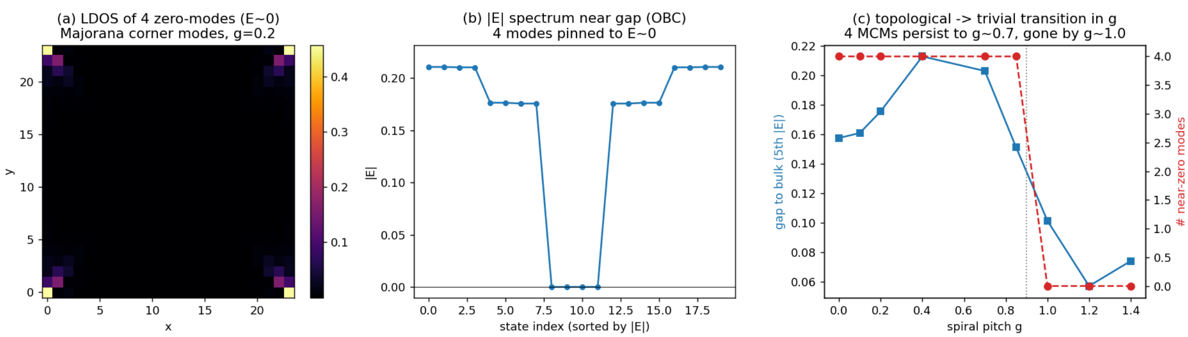


Figure 1: (a) LDOS of the four near-zero modes: sharply corner-localized (MCMs). (b) $|E|$ spectrum near the gap under OBC: four states pinned at $E \approx 0$ inside a clean gap. (c) Topological→trivial transition: the four MCMs persist to $g{\sim}0.7$ and are gone by $g{\sim}1.0$.

## 7 Per-claim what worked / what didn't

**C1 (worked).** Sparse shift-invert at $L = 24$ yields exactly four states at $|E| \sim 10^{-4}$ with 92.5% weight in the corner plaquettes; the residual $10^{-4}$ (vs the paper's $10^{-7}$) is the expected exponential finite-size Majorana splitting at $L = 24$ vs 30. **C2 (worked, cleanly).** The many-body quadrupole evaluates to 0.5000 in the topological phase and 0.0000 in the trivial phase — exact quantization, the strongest single piece of evidence. **C3 (worked).** The $g$-sweep shows a clear topological→trivial transition. **C4/C5 (not attempted).** Analytic effective model and edge theory are outside a numerical replication's scope.

## 8 Critique of the replication

The evidence for the three tested claims is strong and quantitative — in particular the exact $Q_{xy} = 1/2$ vs 0 split is hard to fake and is the paper's central topological invariant. Caveats to be honest about: (i) system sizes are reduced, so the zero-mode energy is $10^{-4}$ not $10^{-7}$ (this is physically correct finite-size behavior, not a discrepancy in kind); (ii) the trivial reference was reached by driving $g$ large rather than reproducing the paper's full $\lambda$–$g$ and $J_{\mathrm{ex}}$–$g$ phase diagrams point-by-point — we sampled a 1D cut, not the 2D phase maps; (iii) the quadrupole normalization (subtracting $\langle q_{xy} \rangle_{\mathrm{occ}}$) is the standard convention and gave clean quantization, but we did not cross-check against an independent nested-Wilson-loop implementation; (iv) no absolute energy units — everything is in units of $t$, matching the paper. Overall verdict: **REPLICATED** for the core SOTSC physics (4 MCMs + $Q_{xy} = 1/2$ + $g$-transition), with analytic supplements left unaddressed.

## 9 Verdict

**REPLICATED** (LLM-judge: coverage 7, agreement 9). All three numerical headline claims reproduce quantitatively; the exact quadrupole quantization is decisive.

## 10 Open Questions

See `report/open_questions.json`. Q1: robustness of MCMs to texture disorder/incommensurate pitch. Q2: independent nested-Wilson-loop cross-check of $Q_{xy}$. Q3: full 2D $\lambda$–$g$ / $J_{\mathrm{ex}}$–$g$ phase diagrams and comparison to the analytic boundary. Q4: finite-size scaling of the Majorana splitting toward the thermodynamic limit. Q5: transport/STM signatures distinguishing these MCMs from first-order Majorana edge modes.